Draft v1 (skeleton): caption once, ask cheaply

Started 2026-10-20; v1 the same evening, after C3–C6; v1.1 on 2026-10-21, after C7–C10; v1.2 on 2026-10-22, with §4–5 (Results, Cost) in prose. The other sections are still a skeleton, not prose. It shows which claims we make, what backs each one, and what is missing. Numbers come from `docs/new_direction/research_log.md` (entry names in brackets) and `results/index_qa/reports/*.txt`. **⟨PENDING: X⟩** marks a number that run X will fill. Terms follow the *Terms* block of `research_plan_caption_index.md`: source, visual budget, captioner and modality effects. CIs are 95% (98.33% for C1's P1–P3); "cl." marks a CI over clip clusters.

Target: ACL 2027 (ARR, January 2027), long paper (8 pages). **G5 is case 1** (C3a positive), so the plan says ACL main; long or short is decided at G9 (Nov 30).

Claims and evidence (working table, not a section of the

paper)

#	Claim (as the paper would state it)	Evidence	Status
1	**Who writes the caption matters.** With the same frames, a stronger model's question-blind captions make a cheap reader more accurate than the reader's own captions.	P3 +0.148 [+0.088, +0.209] (C1, pre-registered); C2a, Gemini at about matched visual tokens, +0.081 [+0.030, +0.131]; **C3a, Sonnet at identical input, +0.064 [+0.017, +0.111], cl. [+0.011, +0.119]**; open reader (C12b, Qwen2.5-VL-7B), Sonnet − Qwen captions +0.125, cl. [+0.070, +0.182]; on minimal pairs, C4b +0.104 [+0.017, +0.190]	**Backed**, in two families and at identical input, three ways. Smaller than predicted (C3a predicted +0.09).
2	**Text itself is not easier for the reader.** The reader's own captions of its frames are no better than the frames, and for a strong reader they are worse.	Haiku: P2 +0.030 [−0.034, +0.094]. Sonnet on its own captions: −0.088 [−0.135, −0.044], cl. [−0.140, −0.036]. Qwen (C12a): −0.121, cl. [−0.170, −0.074]	**Backed.** Haiku gives a bound of ≤ +0.09; Sonnet and Qwen give a cost. Qwen reads frames better than Haiku (0.657 vs. 0.613) and captions worst, so the sign follows frames vs. own captions, not model size.
2a	**The question-blind caption caps accuracy, whatever the reader.** On Sonnet's captions, Haiku and Sonnet score the same.	Haiku 0.707 vs. Sonnet 0.710 on Sonnet captions: −0.003 [−0.047, +0.040] (C3 result, unplanned). Qwen − Haiku on Sonnet captions (C12g, predicted −0.05): −0.047 [−0.094, +0.000]; two-clip −0.29, stacked −0.17	**New, exploratory.** One captioner. Qwen reaches the cap at the boundary, and not where windows must be combined. On Gemini video captions Sonnet leads Haiku by 0.054, so the cap is not reached with every index.
2b	…**except on two-clip videos, where captions help a weak reader** (+0.22 accuracy; +0.36 pair accuracy).	A3; C4a two-clip +0.36 [+0.16, +0.56]; C3b two-clip +0.33 [+0.20, +0.48]	**Backed.** The mechanism is open: describing in one call gives two-clip +0.09 [−0.06, +0.24] (C5a), below the self-captions by −0.13 [−0.30, +0.02] (C5b). This leans to "a separate call per window", without showing it.
3	**The index pays through the reader downgrade, not through fewer tokens.** A cheap reader on a strong captioner's captions costs 1/23 to 1/26 per question and repays the captioning within 1.2–3.7 questions per video. Its accuracy cost depends on the captioner.	Q* 1.2–3.7 (A2, C10, C11); the Claude chain, Sonnet captions once and Haiku answers vs. Sonnet answers: Q* 2.3 [2.2, 2.4], accuracy **−0.091** [−0.145, −0.040]. Gemini frame captions: Q* 3.7, accuracy −0.007 [−0.057, +0.044]. Same-reader savings: 18% (Haiku), 30% (Sonnet). **C9:** Sonnet captioning 448 px frames (720 visual tokens per window) does not close the Claude gap: C9a +0.003 [−0.037, +0.047]; Haiku on them vs. Sonnet on frames −0.088 [−0.138, −0.037], Q* 3.0	**Backed for cost.** "Close to the strong reader" holds only for Gemini frame captions (≈ 5,500 visual tokens per window). The Claude chain loses 9 points, and C9 shows the loss is not the captioner's 224 px frames: it sits on reversed (−0.18) and stacked (−0.24) videos, as in claims 4–5. **Reword: the index trades ≈ 0–9 points for a 23–26× lower query cost.**
4a	*(exploratory, C11)* **Telling the captioner the kinds of question (not the question) helps a little, but not with the stacked comparison.**	C11b +0.040 [+0.003, +0.081] (Sonnet), C11a +0.027 [−0.013, +0.067] (Haiku), both as predicted (+0.03). Sonnet on its own aspect captions vs. its frames −0.037 (C10d −0.077). Against per-	**Weakly backed** (the clustered CI touches 0). Captions are 15% longer, so more words may explain part of it. Reversal mentions do not rise. Supports

#	Claim (as the paper would state it)	Evidence	Status
		window captions: stacked 0.39 vs. 0.39 (Sonnet)	Prop. 4: the stacked loss needs the question itself (C5b).
3a	*(robustness, cost)* **Splitting short videos into 5 s windows costs nothing; one caption per video halves the index cost.**	C10a +0.013 [−0.027, +0.057] (Haiku), C10b +0.010 [−0.030, +0.051] (Sonnet); one-window control ≈ 0. Whole-video index $0.0049 vs. $0.0091 per video; Q* 1.2 [1.2, 1.3] vs. Sonnet on frames, at −0.077	**Backed** (exploratory, predicted +0.00). By type, reversed +0.12 for Haiku, and stacked leans negative; descriptive.
4	**Question-blind captions drop comparisons between things shown at the same time** (stacked clips at two speeds), even from a strong captioner. The loss comes from not seeing the question, not from the text.	C4 stacked pairs: caption arms' pair accuracy 0.10–0.20 vs. Sonnet on frames 0.51; C4c −0.34 [−0.60, −0.08]; C4e −0.41 [−0.62, −0.23]; same-answer rate above Sonnet on frames' in 3 of 4 caption arms. C5b stacked: a description that sees the question beats question-blind self-captions by **+0.22 [+0.09, +0.37]**.	**Backed** (exploratory, written in advance in C4, C5 and C7). The audit (C7) labels 0.93–1.00 of stacked failures "silent" in every arm (predicted ≥ 0.70). Judge vs. hand labels κ = 0.81.
5	**Reversed clips: question-blind captions narrate the forward event, even when the captioner can see the reversal.**	Sonnet on frames beats Haiku on Sonnet captions on reversed videos: C3c −0.19 [−0.32, −0.07]. Sonnet on its own captions −0.18 [−0.30, −0.05] vs. its frames. Sonnet captions do not report reversals more often than Haiku's: C3a reversed +0.01, C4b reversal +0.06 [−0.01, +0.13]. Gemini's do: C2a reversed +0.14. Wrong answers fall to the prior in both modalities (A3, Haiku). **C8, Sonnet:** question-only accuracy on reversed videos 0.18 (Haiku 0.16); 0.59 of its wrong answers on frames and 0.65 on its own captions equal its question-only answer (uniform 0.43).	**Reworded; partly backed** (the old wording, "a stronger captioner fixes it", failed for Sonnet). The audit finds forward narration ("wrong") in 0.28–0.45 of reversed failures, below the predicted ≥ 0.50; the rest are "silent", where the captions do not commit to a direction. Say "narrate forward or don't commit". The fall-back to the forward prior now holds for both readers (C8).
6	**Captions beat frames for Haiku even when the frames get 2× more tokens.**	C6: 448 px frames, 2,676 input tokens, 0.650. Captions minus 448 px: Gemini video +0.088 [+0.037, +0.138]; Sonnet +0.057 [+0.003, +0.111]; Gemini frame +0.141. C6a (448 − 224 px) +0.037 [−0.010, +0.084]	**Backed.** The gain is mostly on two-clip videos.
7	**Caption arms fail mainly by omission.** Most wrong answers come from captions that don't decide the question; few come from the reader.	C7: silent 0.61–0.73 of each caption arm's wrong answers, misstatement 0.15–0.28, reader error 0.05–0.17 (predicted reader error 0.15–0.25)	**Backed by the judge**; κ = 0.81 against 40 hand labels, where every disagreement is a hand "silent", so omission is if anything undercounted. Supports claim 2a (the cap).
8	*(minor)* **Writing before answering helps a cheap reader** (+0.07), at 2× the query cost. It is not an index.	C5a +0.071, cl. [+0.029, +0.114]; $0.000415 vs. $0.000204 per question	Backed. A paragraph in §4.6.

#	Claim (as the paper would state it)	Evidence	Status
—	*Dropped:* "captions lose continuous quantities (direction, speed)"	P1 −0.10 [−0.23, +0.03], not passed; aspect ≈ video type (A3)	Reported as a null in §4.5
—	*Dropped:* "the captioner's knowledge"	No arm separates perceiving, recognizing and writing (D2)	We say "captioner effect"
—	*Dropped:* "a captioner that reports reversals more often fixes reversed clips"	C3a reversed +0.01	Folded into claim 5

Headline (case 1): "Given exactly the reader's frames, a stronger captioner's question-blind captions make a cheap reader more accurate (+0.06 to +0.15). Turning the frames into text does not (≤ +0.09 for Haiku; −0.09 for Sonnet). A caption index lets a 23–26× cheaper reader answer at 0–9 points below the strong reader on frames, and repays itself within 1.2–3.7 questions per video. What it loses is what a question-blind captioner does not write down: the comparison between stacked copies, and reversals."

Tension to resolve while writing: claims 1 and 2 say the captioner adds usable information. Claims 2a, 4 and 5 say question-blindness caps it. Both hold: the captioner helps a reader below the cap (Haiku) and hurts one above it (Sonnet). **Fig. 3** (`docs/paper/figures/caption_index_fig3_readers.png`) shows this: from Haiku to Sonnet, frames rise 0.61 → 0.80 and cross both indexes; Sonnet captions stay at 0.71; Gemini video captions rise 0.74 → 0.79. The CIs in Fig. 3 are per arm; the paired differences (claim 2a, C3c) are tighter.

Abstract (bullets)

- Setting: captions as a one-time, question-independent index of short videos, read later by an LLM.
- Question: when and why does a reader answer better from the index than from frames, and what does it cost?
- Design: an information control. The captioner sees exactly the reader's frames, which splits the caption effect into source, visual budget, captioner and modality effects. Pre-registered on 297 TempCompass questions, with exploratory follow-ups each logged with predictions before running.
- Findings:
  - The captioner effect is +0.15 (P3), +0.08 at matched visual tokens (C2a), and +0.06 within one model family at identical input (C3a).
  - The modality effect is ≤ +0.09 for Haiku and −0.09 for Sonnet.
  - The index pays through the reader downgrade: a 23–26× cheaper query, Q* 1.2–3.7 questions per video, at 0–9 points of accuracy.
- Where captions fail: minimal pairs show that question-blind captions lose the comparison in stacked clips (pair accuracy 0.10–0.20 vs. 0.51) and narrate reversed clips forward. A question-aware description recovers the stacked loss (+0.22). An audit of the wrong answers finds that 61–73% come from captions that are silent on the question, and only 5–17% from the reader.

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1. Introduction (bullets)

- **Motivation.** Captioning a video is paid once; questions come later and repeatedly. Caption-then-reason pipelines are widely used (LLoVi, VideoTree, SiLVR), and "caption once, reuse" is now published for long video (CFD, CapMem, Sprout). Two questions are open:
    1. When the captions win, **why**? Did the captioner see more, spend more visual tokens, write differently, or is text easier to read than pixels?
    2. **Does it pay**, counting the captioning and the reader's full per-question cost?
- **Approach.** The information control (Prop. 1): give the captioner exactly the frames the reader sees. Then any caption advantage is usable information added by the captioner or by the modality, not information about the video.
- **Contributions:**
    1. A decomposition of the caption effect into source, visual budget, captioner and modality effects, with arms that change one thing each, in two model families (§2.3, Table 2).
    2. A pre-registered test plus a within-family replication at identical input: the captioner effect is positive, and the modality effect is ≤ +0.09 (and negative for a strong reader) (§4.1–4.2).
    3. A cost account per question (input, output and thinking tokens, list-price dollars) and the break-even Q* (§5).
    4. What question-blind captions lose, shown with minimal pairs, a question-aware control and a caption audit (§4.3–4.4).
- **Scope:** short videos (≈ 10 s), multiple choice, one benchmark.

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2. Setup

2.1 Task and data

- **TempCompass** multiple choice (CC BY-NC). Aspects: action, speed, direction, order, attribute change. Median video 10 s.
- **Development set:** 100 questions (K0–K2), used to set the hypotheses.
- **Confirmatory set:** 300 questions sampled (60 per aspect, one per video, on videos the development set does not use), 297 after Amendment 2; pre-registered (C1).
- **Video types** (A3): plain (137), reversed (74), stacked (41), two-clip (45). They nearly determine the aspect. There are 156 clusters of videos that share a filmed clip.
- **Minimal pairs** (C4): 193 confirmatory questions, each re-asked with the same wording on a sibling video whose correct option differs (107 reversal, 41 stacked, 45 two-clip). Chance pair accuracy is 0.093.
- **Windows:** 5 s; frames at 1 fps ($t = 5p + k$), 224 px long edge (40 Claude image tokens each).
    - Videos are short: median 9.7 s, 1–7 windows (1: 16%, 2: 42%).
    - The window length was checked: one caption per video from the same frames does no better (C10, claim 3a).
    - The per-window design is kept because it is what scales to long video.

2.2 Arms

Arm	Reader sees	Visual tokens for the reader / captioner per 5 s window
Question only	question and options	0
1 frame	the middle frame of each window	40
5 frames	all 5 frames of each window	200
5 frames, 448 px	the same frames at 448 px	720
Describe, then answer	5 frames; writes one sentence per window, then answers	200
Self-captions	the reader's own isolated caption of each window's 5 frames	— / 200
Sonnet captions	Sonnet's isolated caption of the same 5 frames	— / 200
Sonnet captions, 448 px	the same, from the 448 px frames (C9)	— / 720
Sonnet whole-video captions	one Sonnet caption of all of a video's frames, 45 words per window (C10)	— / 200
Sonnet aspect-aware captions	the same, with one prompt bullet naming the kinds of question to come (C11)	— / 200
Gemini frame-clip captions	Gemini's caption of the same 5 frames, played as a 1 fps clip	— / ≈ 280–330
Gemini frame captions	Gemini's caption of the same 5 frames, as images	— / ≈ 5,500
Gemini video captions	Gemini's caption of the video window	— / ≈ 330

- **Readers:** Claude Haiku 5.5 and Sonnet 5.5 (pinned IDs, effort medium), through `claude -p` with no tools, isolated per call.
- **Captioners:** Haiku (self), Sonnet, Gemini 3.8 Flash. Every caption is question-blind and window-isolated, and all use one prompt.

2.3 The four effects (Haiku reads)

Effect	Arms compared	Claude chain	Gemini chain
Source	video captions − frame-clip captions	0 by design	C2c +0.013 [−0.030, +0.061]
Visual budget	frame captions − frame-clip captions	0 by design	C2b +0.067 [+0.030, +0.108]
Captioner	other captioner − self-captions, same pixels	C3a +0.064 [+0.017, +0.111]	C2a +0.081 [+0.030, +0.131]
Modality	self-captions − 5 frames	P2 +0.030 [−0.034, +0.094]	(same)
Total	other captioner − 5 frames	C3b +0.094 [+0.044, +0.148]	P3 + P2; frame captions − 5 frames +0.178

- C3d (Sonnet − Gemini frame-clip captions) is −0.017 [−0.071, +0.037]. At about equal visual tokens the two families' captioners are equally useful to Haiku.

- **Visual budget within Claude (C9a):** Sonnet's captions of 448 px frames − of 224 px frames, +0.003 [−0.037, +0.047]. Unlike Gemini's (C2b +0.067), Claude's captioner gains nothing from 3.6× the visual tokens. The two budget changes differ in size (200 → 720 vs. ≈ 300 → 5,500), so say "in this range".

2.4 Statistics and pre-registration

- C1 was pre-registered: P1–P3 at 98.33% CIs (Bonferroni over three); secondary estimates at 95%.
- C2–C7 are exploratory. Each was logged with predictions before its first call, and each analysis was run once.
- Paired bootstrap over questions; clip-clustered bootstrap as a check (the CIs agree everywhere; report both or state it).
- **Many exploratory terms** (≈ 30 across C2–C6). The paper should say which were predicted directionally and report all of them in the appendix.
- **Missing:** a public, timestamped archive of the pre-registration commit (plan §5 step 9; G11).

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3. Theory (short)

- **Prop. 1 (garbling bound).** A caption made from the reader's own frames cannot add information about the video, for an unbounded reader. For a fixed, bounded reader, it can add *usable* information (Xu et al. 2020), and that is what the captioner effect measures.

    - **Retention**, (captions − question only) / (frames − question only), is the share of the frames' gain the index keeps. CIs are over clip clusters throughout. For Sonnet (C8, D3): Gemini video captions 0.98 [0.85, 1.13]; its own captions 0.77 [0.64, 0.90]. For Haiku, every index's point estimate exceeds 1 because Haiku on frames is below what the captions hold (claim 2a). The CIs exclude 1 for every index but its own captions (D3, over clip clusters):
    - self-captions 1.15 [0.91, 1.46];
    - Sonnet captions 1.47 [1.18, 1.95];
    - Gemini video captions 1.62 [1.33, 2.10];
    - Gemini frame captions 1.88 [1.53, 2.51].

    The variants (appendix): Haiku retains 1.48 on Sonnet's 448 px captions, 1.53 on its whole-video captions and 1.55 on Gemini frame-clip captions. Sonnet retains 0.82 and 0.80 on its own 448 px and whole-video captions.

- **Corollary.** The total caption effect = source + captioner + modality, plus a visual budget term when the visual tokens differ.

- **Prop. 2 (break-even).** $Q^* = C(\tau) / (c_f - c_\tau)$: the captioning cost over the per-question saving. With a reader downgrade, c_τ is the cheap reader's cost.
- **Prop. 4 (price of being question-blind)** is now used. C5b stacked (+0.22) and claim 4 are its evidence: a question-aware description recovers what the question-blind caption omits.
- **Possible addition:** the cap of claim 2a. The accuracy from a question-blind caption is bounded by what the caption holds, so readers above that bound lose and readers below it gain. This is informal; decide whether it needs a proposition.

- Prop. 3 (routing headroom) is not used.

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4. Results

Prose v1 (2026-10-22). For review, the headings name the claim (claims table) each subsection backs; the tags go before submission.

All results are on the 297 confirmatory questions, and no question is missing from any arm. P1–P3 were pre-registered (C1) and carry 98.33% CIs. Every other comparison is exploratory: it was written down with a predicted value before its first call, and its analysis was run once. These carry 95% paired bootstrap CIs over questions. "cl." marks a CI from resampling the 156 clusters of videos that share a filmed clip; the two kinds of CI agree throughout (Appendix D). Differences are in accuracy, so +0.064 is 6.4 points.

Table 1 lists every arm. Two rows anchor the rest. With the question and options alone, both readers score 0.41 (Haiku 0.411, Sonnet 0.414; difference +0.003 [−0.061, +0.064]), well above chance (≈ 0.30): the wording carries a strong prior, and the two readers share it. Given five frames per window, Sonnet scores 0.185 more than Haiku (0.798 vs. 0.613). Since the floor is shared, that whole gap comes from reading the input. Every index Haiku reads lands between these two points (0.643–0.791), and every index Sonnet reads lands below its frames (0.710–0.791).

Table 1. Accuracy and list-price cost per question (297 questions). Input tokens are the median per question and include the fixed reader prompt (≈ 1,025 tokens). $ per question includes output and thinking tokens. Index $ per video is the one-time captioning cost, averaged over videos (§5).

Reader, input	Acc.	Input tok. (median)	$ / question	Index $ / video
Haiku, question only	0.411	1,025	0.000109	—
Haiku, 1 frame	0.582	1,189	0.000146	—
Haiku, 5 frames	0.613	1,520	0.000204	—
Haiku, 5 frames 448 px	0.650	2,676	0.000339	—
Haiku, describe then answer	0.684	1,579	0.000415	—
Haiku, self-captions	0.643	1,283	0.000189	0.00058
Haiku, Sonnet captions	0.707	1,280	0.000181	0.00910
Haiku, Gemini frame-clip captions	0.724	1,232	0.000167	0.00760
Haiku, Gemini frame captions	0.791	1,229	0.000167	0.01461
Haiku, Gemini video captions	0.737	1,236	0.000168	0.00726
Haiku, Sonnet captions 448 px	0.710	1,279	0.000180	0.01167
Haiku, Sonnet whole-video captions	0.721	1,194	0.000156	0.00492
Haiku, Sonnet aspect-aware captions	0.747	1,213	0.000162	0.00527
Sonnet, question only	0.414	1,027	0.002182	—
Sonnet, 5 frames	0.798	1,522	0.004112	—
Sonnet, Sonnet captions	0.710	1,282	0.003089	0.00910
Sonnet, Sonnet captions 448 px	0.731	1,281	0.003103	0.01167
Sonnet, Sonnet whole-video captions	0.721	1,196	0.002849	0.00492
Sonnet, Sonnet aspect-aware captions	0.761	1,215	0.002897	0.00527
Sonnet, Gemini video captions	0.791	1,238	0.002868	0.00726

4.1 Who writes the caption matters (claim 1)

Table 2 (§2.3) splits the caption effect into source, visual budget, captioner and modality effects. The pre-registered test is P3. With Haiku as the reader, Gemini's captions of exactly the reader's frames beat Haiku's own captions of them by +0.148 [+0.088, +0.209], 2.5× the predicted +0.06. The gain is positive in every aspect (+0.10 to +0.20) and in every video type (Appendix D).

P3 compares two captioners that spent different visual tokens on the same pixels. Gemini's image path encodes each 224 px frame at ≈ 1,100 tokens, ≈ 5,500 per window, against Haiku's 200. Captioning the same frames as a 1 fps clip through Gemini's video path (≈ 280–330 tokens per window, equal to the video arm's in 693 of 761 windows) splits P3 into two halves:

- a captioner effect at about matched visual tokens, C2a = +0.081 [+0.030, +0.131], cl. [+0.035, +0.125];
- a visual budget effect, C2b = +0.067 [+0.030, +0.108].

The source effect is near zero (C2c +0.013 [−0.030, +0.061]): with captioner and visual tokens fixed, captions of the original video are no more useful than captions of the reader's 224 px frames. Gemini's video path reduces each frame to ≈ 66 tokens, so the source's resolution mostly does not reach the captioner. All four C2 predictions were within 0.02 of the results.

Within one model family, the match is exact. Sonnet captions the same five JPEGs per window, with the same prompt and effort as Haiku's self-captions. Every Claude model counts image tokens by the same rule, and Sonnet's caption calls took exactly 2 more input tokens than Haiku's in every window (876). Only the captioning model differs. Haiku reads Sonnet's captions more accurately than its own: **C3a = +0.064 [+0.017, +0.111]**, cl. [+0.011, +0.119]. The predicted +0.09 lies inside the CI. At about equal visual tokens, the two families' captioners are equally useful to Haiku (Sonnet − Gemini frame-clip captions, C3d −0.017 [−0.071, +0.037]). On minimal pairs (§4.3), where a guess cannot score, the captioner effect is +0.104 [+0.017, +0.190] in pair accuracy (C4b).

Unlike Gemini, Claude's captioner gains nothing from a larger visual budget in the range we tested. Sonnet's captions of 448 px frames (up to 720 visual tokens per window, 3.6× as many) are no more useful to Haiku than its captions of the 224 px frames: C9a +0.003 [−0.037, +0.047], against a predicted +0.03. With power ≈ 0.25 for +0.03, this is weak evidence against a small effect, but the CI rules out more than about +0.045. The two budget changes also differ in size (200 → 720 tokens here, ≈ 300 → 5,500 for Gemini).

4.2 Text itself is not easier for the reader (claims 2, 2a, 6)

The modality effect asks whether a reader does better on its own words than on the pixels they came from. For Haiku this is the pre-registered P2: self-captions − 5 frames = +0.030 [−0.034, +0.094]. Both the prediction (−0.08) and any gain above +0.09 lie outside the CI; IsoBench, by contrast, finds 15–29 points for exact text renderings of images. For Sonnet, its own captions cost accuracy: −0.088 [−0.135, −0.044], cl. [−0.140, −0.036], and −0.067 [−0.114, −0.024] when it captions the 448 px frames (C9c). The modality effect changes sign with the strength of the reader.

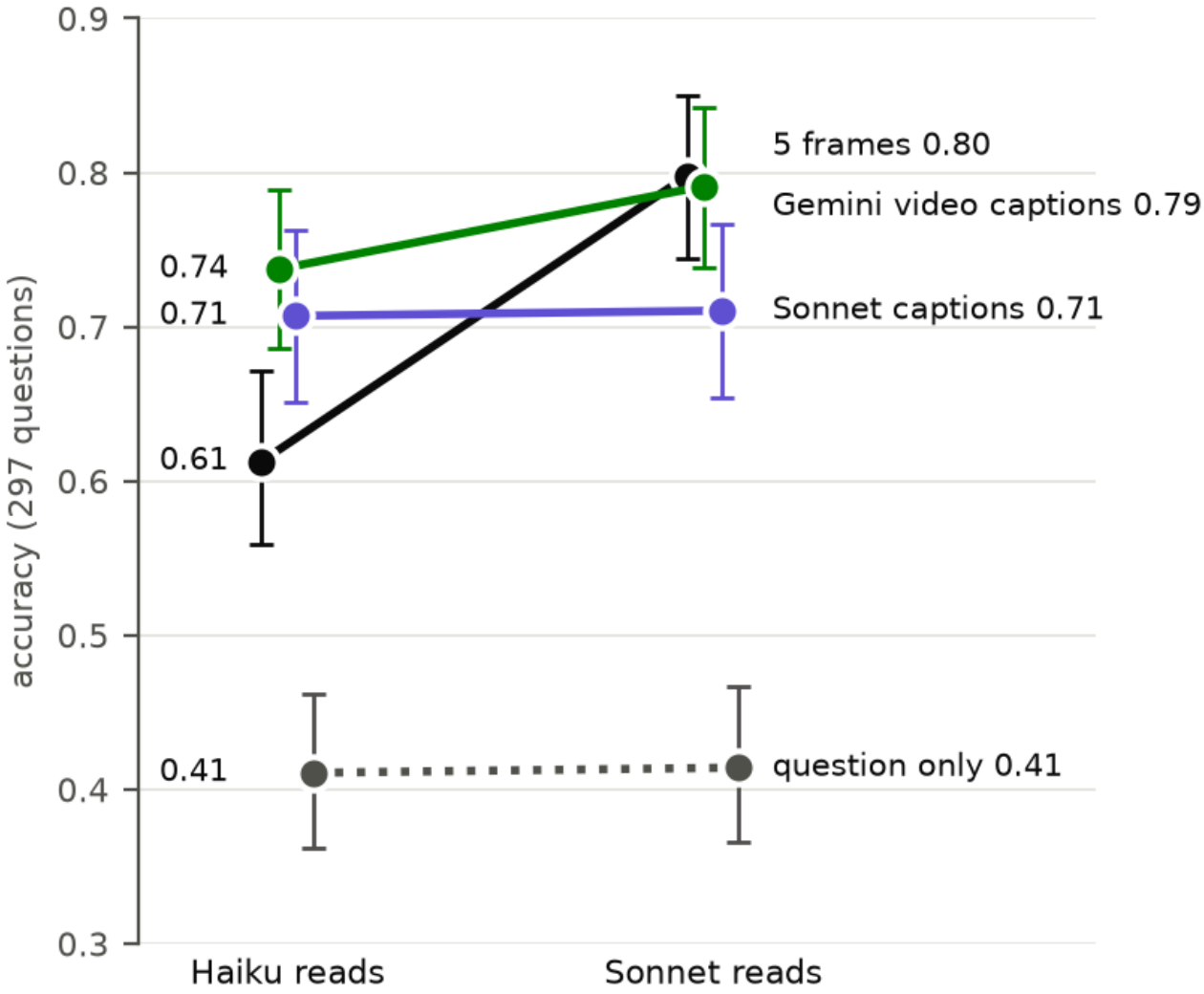


Figure 3. Accuracy by reader for the question alone, 5 frames, and the two indexes both readers read. 95% CIs over clip clusters. The frames line crosses both index lines.

Figure 3 shows why. From Haiku to Sonnet, accuracy on frames rises from 0.613 to 0.798, while accuracy on Sonnet's captions stays flat (0.707 → 0.710; paired difference −0.003 [−0.047, +0.040]). Once the video is in these captions, a reader that costs 17× less per question loses nothing, and a stronger reader gains nothing. We read this as a cap: a question-blind caption bounds any reader's accuracy by what the caption holds. Haiku on frames is below the cap of Sonnet's captions and gains from them; Sonnet is above it and loses. The cap depends on the index. Sonnet reads Gemini's video captions better than its own (0.791 vs. 0.710, +0.081 [+0.034, +0.128]), and keeps 0.98, cl. [0.85, 1.13], of its frames' gain over the question alone on them, against 0.77, cl. [0.64, 0.90], on its own captions (retention, §3). This pattern was not predicted: it appeared in C3, and it is exploratory.

Haiku's advantage on captions does not come from starving the frames. At 448 px, Haiku's frames take 2,676 input tokens per question, 1.76× the 224 px arm, and gain +0.037 [−0.010, +0.084] (C6a). Captions at ≈ 1,280 input tokens still beat them: Gemini frame captions by +0.141 [+0.088, +0.195], Gemini video captions by +0.088 [+0.037, +0.138], and Sonnet captions by +0.057 [+0.003, +0.111]. The advantage sits on two-clip videos (+0.31 to +0.38), and for Gemini frame captions also on plain and reversed ones. On stacked videos, every caption arm is within ±0.07 of the 448 px frames.

4.3 Where captions lose: video types and minimal pairs (claims 2b, 4, 5)

TempCompass builds most of its videos to defeat single-frame and wording shortcuts (§2.1): reversed clips, stacked copies of one clip at two speeds, and two clips in sequence or side by side. Built videos are 54% of the confirmatory set. The video type nearly determines the aspect (every stacked question is about speed, and every two-clip question about order), so we report by type. Figure 2 shows accuracy by type. On plain videos, the five arms lie within 0.09 of each other (0.80–0.89). They separate on the built videos.

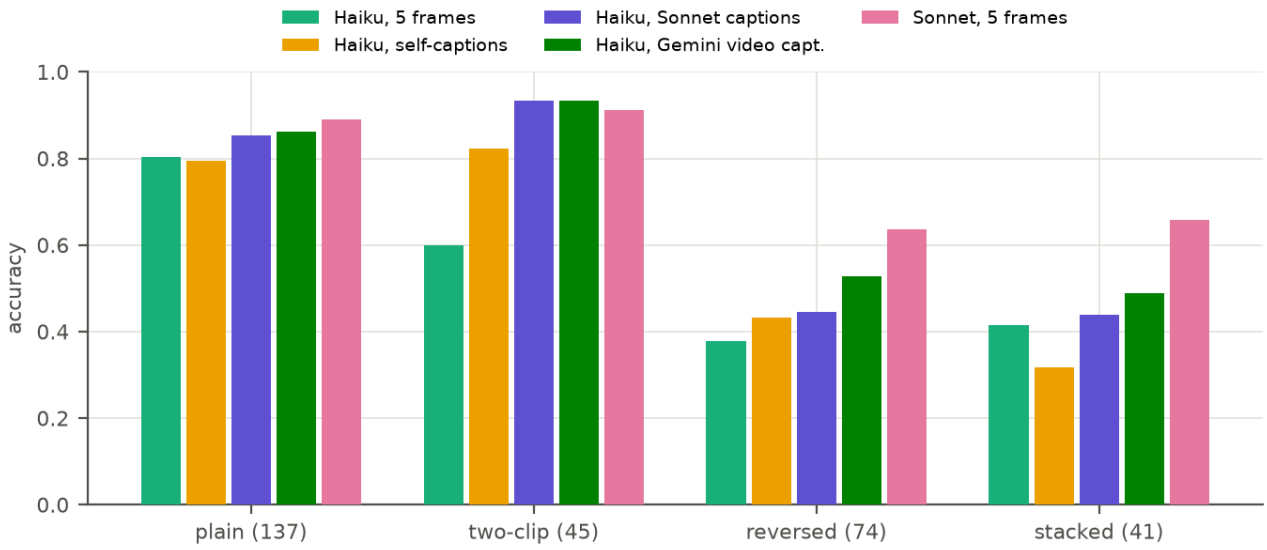


Figure 2. Accuracy by video type for five arms (297 questions; n per type in brackets).

Accuracy on one video cannot tell a representation that lost the fact from a reader that guessed. Minimal pairs can (C4). Each of 193 confirmatory questions is asked again, with the same wording and option order, on a sibling video built from the same filmed clip whose correct option differs:

- the plain and the reversed version of a clip (107 pairs);
- another stacked version at other speeds (41);
- another arrangement of the same two clips (45).

An arm that does not receive the fact gives the same answer on both videos, and gets at most one of them right. Table 3 reports pair accuracy, the share of pairs answered correctly on both sides (chance 0.093), and the same-answer rate.

Table 3. Minimal pairs (C4): pair accuracy by group, and the share of pairs with the same answer on both videos.

Arm	All	Reversal (107)	Stacked (41)	Two-clip (45)	Same answer
Haiku, 5 frames	0.233	0.28	0.17	0.18	0.54
Haiku, self-captions	0.326	0.33	0.10	0.53	0.42
Haiku, Sonnet captions	0.430	0.38	0.17	0.78	0.43
Haiku, Gemini video captions	0.482	0.47	0.20	0.78	0.37
Sonnet, 5 frames	0.606	0.56	0.51	0.80	0.28
Sonnet, Sonnet captions	0.430	0.40	0.10	0.80	0.39

Two-clip: captions help a weak reader. From frames, Haiku must keep about ten small frames from two clips apart, and on two-clip pairs it is near chance (0.18). Its own captions raise pair accuracy to 0.53, a modality effect of +0.36 [+0.16, +0.56] (C4a); in accuracy, the modality effect on two-clip videos is +0.22 [+0.10, +0.35] (cl.). Sonnet's captions raise Haiku's accuracy over its frames by +0.33 [+0.20, +0.48] (C3b, cl.). Both directions were predicted. The mechanism is open. When Haiku describes each window before answering, in the same call, it gains only +0.09 [−0.06, +0.24] on two-clip videos (C5a), and stays below its self-captions (−0.13 [−0.30, +0.02]). This leans toward the separate caption call per window as the cause, without showing it.

Stacked: question-blind captions drop the comparison. A stacked video shows one clip twice, one copy above the other, at the same or different speeds, and the question asks which copy is faster. Sonnet on frames answers 0.51 of stacked pairs correctly on both sides. Every caption arm scores 0.10–0.20, near pair chance, and that includes Sonnet reading its own captions (0.10). Sonnet's pair accuracy on its own captions is 0.41 below its frames (C4e, [−0.62, −0.23]), and Haiku on Sonnet's captions is 0.34 below Sonnet on frames (C4c, [−0.60, −0.08]). Three of the four caption arms give the same answer on both videos of a pair more often than Sonnet on frames does (for example, Haiku on Sonnet captions +0.32 [+0.07, +0.55]). The captions do not say how the copies differ. For a man jumping rope in two copies (answer: the man above is faster), Gemini writes "An athletic man … jumps rope in a split-screen video", and Haiku's own captions describe one woman skipping rope. Sonnet sees the difference when it looks with the question in view, and loses it when it writes without the question. So the loss lies in what a question-blind captioner chooses to write.

Reversed: captions narrate forward, or do not commit. On reversed videos, the question alone points to the forward event: both readers score below chance on it (Haiku 0.16, Sonnet 0.18). Sonnet reading the frames recovers part of the reversal (0.64), but its question-blind captions carry little of it. Haiku on Sonnet's captions trails Sonnet on frames by −0.19 [−0.32, −0.07] (C3c, cl.), and Sonnet on its own captions trails its frames by −0.18 [−0.30, −0.05] (cl.). A stronger captioner does not help here. Sonnet's captions report reversals no more often than Haiku's (C3a on reversed videos +0.01 [−0.08, +0.10]; C4b on reversal pairs +0.06 [−0.01, +0.13], where a positive effect was predicted). Gemini's captions do report them more often (C2a on reversed videos +0.14 [+0.04, +0.24]). When a reader misses the reversal, it falls back to the prior. Among Sonnet's wrong answers on reversed videos, the share equal to its question-only answer is 0.59 with frames (27 wrong) and 0.65 with its own captions (40), against 0.43 if errors were uniform; for Haiku it is 0.67 with frames and 0.55 with self-captions. For a pile of salt played backwards (answer: shrinking), all five of Gemini's video captions describe salt pouring onto the pile, and Haiku's own captions say it "grows taller".

4.4 Why caption arms fail: an audit (claim 7)

For every wrong answer from captions in seven reader arms (442 items), a judge (Sonnet) saw the captions, the question, the options and the correct answer. It labelled the captions as stating the correct option (so the error is the reader's), stating a wrong option (a misstatement), or deciding neither (an omission), and quoted the caption as evidence (C7; prompt in Appendix A). Table 4 gives each label's share of each arm's wrong answers.

Omission dominates: 0.61–0.73 of every arm's wrong answers come from captions that do not decide the question. Misstatements are 0.15–0.28, and reader errors only 0.05–0.17, below the predicted 0.15–0.25. Once a fact is missing from the index, no reader can recover it; this is the cap of §4.2. By video type, the stacked prediction held: in every arm, 0.93–1.00 of stacked failures are omissions (predicted ≥ 0.70). The reversed prediction failed. Misstatement, that is

forward narration, is 0.28–0.45 of reversed failures, below the predicted ≥ 0.50, and omission is 0.45–0.62. On reversed videos, captions narrate the forward event or do not commit to a direction.

The judge also wrote one of the caption sets it labels. As a check, one author labelled 40 items, stratified by the judge's label and blind to it. Agreement is 0.88, $\kappa = 0.81$ [0.64, 0.96], above the threshold of 0.6 set in advance, below which only the hand-labelled sample would have been reported. The judge agrees on all 22 items the author labelled "correct" or "wrong". All 5 disagreements are items the author left "silent" and the judge decided, so if anything Table 4 undercounts omission.

Table 4. Caption audit (C7): share of each reader arm's wrong answers by the judge's label of the captions. Reader error = the captions state the correct option; misstatement = a wrong option; omission = neither.

Reader arm	Reader error	Misstatement	Omission	n wrong
Haiku, self-captions	0.08	0.28	0.63	106
Haiku, Sonnet captions	0.13	0.26	0.61	87
Haiku, Gemini frame-clip captions	0.17	0.15	0.68	82
Haiku, Gemini frame captions	0.10	0.18	0.73	62
Haiku, Gemini video captions	0.10	0.21	0.69	78
Sonnet, Gemini video captions	0.05	0.24	0.71	62
Sonnet, Sonnet captions	0.08	0.26	0.66	86

By video type (Appendix D): two-clip failures are few (1–8 per arm) and mostly reader errors, where the captions state the order and the reader still answers wrongly. Plain failures are 0.60–0.80 omission.

4.5 From question-blind to question-aware (claims 8, 4a)

Two arms move the captioner toward the question. Neither changes the confirmatory captioner prompt.

Describe, then answer (C5). Haiku sees the 5 frames per window and the question, writes one sentence per window, and then answers, all in one call. It scores 0.684, +0.071 over the frames arm, cl. [+0.029, +0.114] (predicted +0.03). The gain is on reversed and stacked videos (+0.12 each). On stacked videos, this question-aware description beats the question-blind self-captions by **+0.22 [+0.09, +0.37]**. So the stacked loss is an omission caused by not seeing the question, not a limit of text (Prop. 4). This arm is not an index: it reads the frames at every question, and it costs 2× the frames arm per question ($0.000415 vs. $0.000204), mostly in thinking tokens.

Telling the captioner the kinds of question (C11). Between question-blind and question-aware sits a task-aware index. Sonnet's whole-video captions (§4.6) get one added prompt bullet that names the five kinds of question to come (actions, speed, direction, order, attribute changes), but not how the benchmark builds its videos. Sonnet reading these captions gains +0.040 [+0.003, +0.081], cl. [+0.000, +0.081], over the whole-video captions, and Haiku +0.027 [−0.013, +0.067]; both match the predicted +0.03. This halves Sonnet's gap to its own frames (−0.077 → −0.037 [−0.081, +0.007]). It does not fix the stacked comparison. Against the per-window captions, Sonnet's stacked accuracy is unchanged (0.39 vs. 0.39), and Sonnet on frames stays 0.27 ahead. Mentions of reversal did not rise (4 of 74 reversed videos, against 6 without the

bullet). The captions are 15% longer at the same length limit, so part of the gain may come from more words rather than the right words. Knowing the task buys a little; the stacked comparison needs the question itself (§8.1).

4.6 Robustness checks and nulls (claim 3a)

Window length (C10). Our videos are short (median 9.7 s; 58% have one or two windows), so we checked the per-window design against one Sonnet caption per video, from the same frames, with the length limit summed over the windows. Whole-video captions are no better: +0.013 [−0.027, +0.057] with Haiku reading, and +0.010 [−0.030, +0.051] with Sonnet (predicted +0.00 and +0.01). Splitting into 5 s windows thus costs at most ≈ 0.05–0.06, and helps by at most ≈ 0.03. On one-window videos, where the two captioning calls are identical, the difference is ≈ 0 for both readers (+0.02 and −0.02), as it should be. One by-type pattern was not predicted: Haiku gains +0.12 [+0.03, +0.23] on reversed videos from whole-video captions, while stacked videos lean the other way for both readers (−0.07 and −0.12; CIs include 0). The whole-video index costs about half as much (§5).

Resampling noise. On the 48 one-window videos, the captioning call is identical in both arms. None of the 48 caption pairs is identical, and the reader's answer changes on 10% (Haiku) to 17% (Sonnet) of those questions. That is as often as removing the split changes it on videos with two or more windows (17%). Per question, a caption is one sample, so comparisons between caption arms need many questions. The by-type terms above (41–137 questions per type) are noisy for that reason.

Aspects (P1, null). The pre-registered aspect contrast, that captions lose direction and speed but keep the discrete aspects, did not pass: −0.101 [−0.231, +0.028] for Sonnet, half the predicted −0.20. Speed is the only aspect where captions lose consistently (−0.10 for Sonnet). Aspect nearly coincides with video type, so the by-aspect tables are in Appendix D.

Thinking (A1, not predicted). At effort medium, the model decides per call whether to think, so thinking is observed, not set. On Gemini's captions, thinking marks wrong answers: accuracy is 0.84–0.90 on calls without thinking and 0.62–0.65 on calls with it. For Haiku on frames it does not (0.61 both ways). Sonnet thinks 3.3× less on captions than on frames.

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5. Cost and break-even (claim 3)

Accounting. We price every call at list price. Claude Haiku 5.5 costs $0.10 / $0.50 and Sonnet 5.5 $2 / $10 per million input / output tokens, with thinking billed as output; computed from the logged tokens, this equals the CLI's reported cost on every call. Gemini captioning is priced at $0.50 / $3.00, thinking included. A reader's cost per question, c, is the mean over the 297 questions. The index cost of a video, $C(\tau)$, is the cost of captioning all its windows once. By Prop. 2, an index τ that a reader answers from at c_τ per question repays its captioning, against a frame arm at c_f, after $Q^* = C(\tau) / (c_f - c_\tau)$ questions per video. CIs on Q^* resample questions, and so videos, since each video has one question.

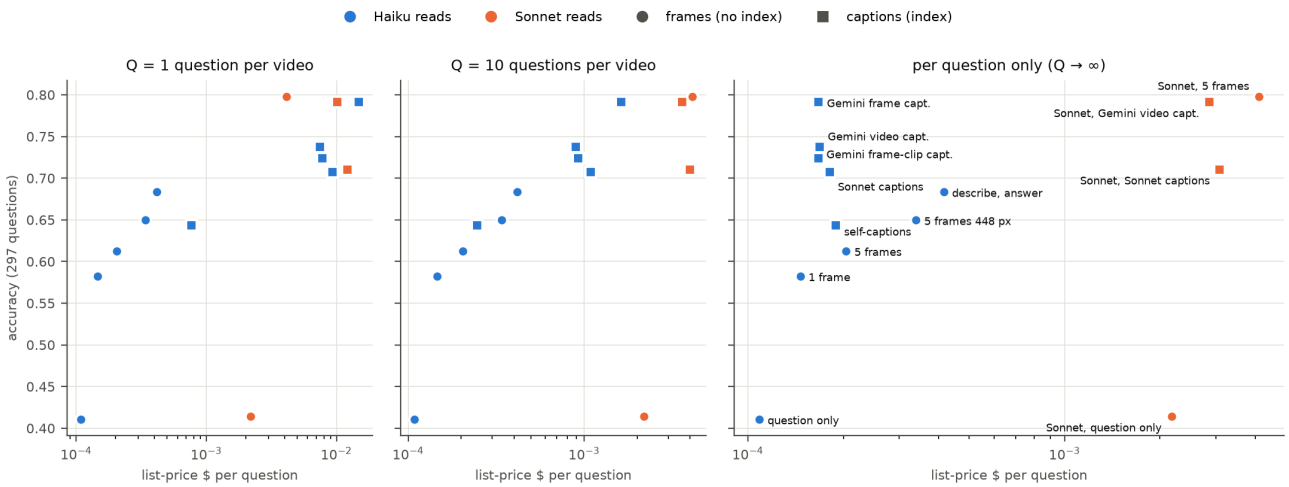


Figure 1. Accuracy against list-price $ per question (log scale), one point per arm, with the index cost spread over Q = 1, 10 and infinitely many questions per video. Circles are frame arms, and squares are index arms. The 448 px caption, whole-video and aspect-aware variants are in Table 1 only.

For one reader, the index saves little. Against 5 frames, Gemini video captions save 18% per question for Haiku and 30% for Sonnet. The fixed reader prompt (≈ 1,025 tokens) dominates the input: captions cut the video-content tokens to 43% of the frames', but the input tokens only to 81%. Sonnet also thinks 3.3× less on captions. So for one reader the index repays slowly. Sonnet on Gemini video captions repays after Q* = 5.8 [5.2, 6.5], at equal accuracy (−0.007 [−0.054, +0.040]). Sonnet on its own captions repays after 8.9 [7.8, 10.2], at −0.088. Haiku on Gemini video captions repays only after 202 [156, 279], though it is also 0.125 more accurate than on its frames. Haiku on Sonnet captions is +0.057 more accurate than Haiku on 448 px frames and cheaper per question, and repays after 57.5 [53.3, 62.1].

The reader downgrade is what pays. Per question, Haiku on any index written by a stronger captioner costs 1/23 to 1/26 of Sonnet on frames (Table 1). The index then repays its captioning within 1.2–3.7 questions per video (Table 6). At Q = 10 questions per video (Fig. 1, middle), every index Haiku reads costs less than half of Sonnet on frames, captioning included. What the downgrade costs in accuracy depends on the index:

- **Gemini frame captions** let Haiku match Sonnet on frames (−0.007 [−0.057, +0.044]). They are the most expensive index ($0.0146 per video), because Gemini's image path spends ≈ 5,500 visual tokens per window.
- **The Claude chain**, where Sonnet captions once and Haiku answers, is at −0.091 [−0.145, −0.040], with Q* = 2.3.
- **One caption per video** instead of one per window halves the index cost ($0.0049 vs. $0.0091 per video) at the same accuracy (§4.6), and Q* falls to 1.2.
- **Aspect-aware captions** (§4.5) bring the Claude chain to −0.051 [−0.104, +0.000] at Q* = 1.3.

In short, a caption index trades 0–9 points of accuracy against the strong reader on frames for a 23–26× cheaper query, and repays itself within 1.2–3.7 questions per video.

Other costs. Latency does not favour captions: the median call on captions is 0.1–1.1 s slower than on frames with the same reader (CLI wall clock with start-up, not controlled for time of day). Thinking is 2–5% of reader tokens.

Table 6. Reader downgrade: Haiku on an index against Sonnet on 5 frames. Q* is the number of questions per video after which the index costs less in total (95% CI over questions).

Index	Q* (questions per video)	Accuracy difference
Gemini video captions	1.8 [1.7, 2.0]	−0.061 [−0.111, −0.010]
Gemini frame-clip captions	1.9 [1.8, 2.1]	−0.074 [−0.125, −0.024]
Sonnet captions	2.3 [2.2, 2.4]	−0.091 [−0.145, −0.040]
Sonnet captions, 448 px	3.0 [2.8, 3.1]	−0.088 [−0.138, −0.037]
Sonnet whole-video captions	1.2 [1.2, 1.3]	−0.077 [−0.128, −0.027]
Sonnet aspect-aware captions (C11)	1.3 [1.3, 1.4]	−0.051 [−0.104, +0.000]
Gemini frame captions	3.7 [3.5, 4.0]	−0.007 [−0.057, +0.044]

Review notes on §4–5 (delete after review)

Decided 2026-10-26 (the user accepted the recommendations): notes 1, 5 and 6 are applied; 2 and 4 wait until the main-text cut; 3 is done at assembly. **2026-10-27: notes 2, 3 and 4 are applied in `acl2027/caption_index.tex`** (main text cut to 8 pages; the Q* table moved to the appendix), which is now the paper's source of truth.

1. ~~**Headline numbers moved with C10 and C11.**~~ Applied: the claims table's headline, claim 3 and the abstract now say 23–26× and Q* 1.2–3.7 (1/23 to 1/25 for the per-window indexes; 1/22 on Haiku's self-captions).
2. **C11 is now reported in §4.5**, so §8.1 can keep the interpretation (the question-blind → task-aware → question-aware gradient) and drop the repeated numbers. *Waits for the main-text cut.*
3. **Table numbers:** the Q* table is Table 6, but it appears before Table 5 (§8.1). Renumber at assembly.
4. **Length:** §4–5 are about 3,500 words (without the tables), with 4 tables and 3 figures, which is more than an 8-page paper has room for. Candidates for the appendix: Table 1's variant rows (448 px captions, whole-video, aspect-aware, frame-clip), the thinking paragraph, and the per-arm by-type audit numbers. *Waits for the main-text cut.*
5. ~~**Hand labels.**~~ Applied: the text keeps "one author labelled 40 items … blind to [the judge]"; the note that Claude pointed out caption details on 6 items (κ unchanged with first-instinct labels) is listed under Appendix C.
6. ~~**Retention CIs.**~~ Applied: retention CIs are over clip clusters everywhere (§3, §4.2, §8), the project's standard for clustered data.

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6. Related work (bullets; literature pass done, log N2 and `related_work_caption_index.md` §10: no scoop on Q1–Q7)

- Caption-then-reason for video QA: LLoVi (the captioner matters: oracle captions +10.8), VideoTree, SiLVR, Vamos (text competitive with visual embeddings; verified).
- Caption indexes and memories: CFD, CapMem, Sprout, VidCtx (question-aware captions +1–2 points; compare with C5b stacked +0.22).
- **A captioner in front of a fixed reader (closest to claim 1):** Prism (images; a 2B captioner + GPT-3.5 ≈ 10× larger VLMs), MetaCaptioner, CapRL / CapRL++ (caption utility as the RL reward). **What none has:** a self-caption control, matched visual tokens, or the

reader's own view of the frames (N2a). That is our gap; word claim 1 as the controlled decomposition.
- Text vs. pixels for the same content: IsoBench (text +15–29 points when the text is exact), TokenSwap. Our modality effect is ≤ +0.09, and negative for the strong reader.
- **A model reading its own description (closest to claims 2 and 2a; images only):**
  - Caption This, Reason That (NeurIPS 2025): own captions instead of the image give a weak 7B model +22 to +29 on synthetic location tasks, and GPT-4o +8 (not significant).
  - GEASS: the model's own caption added to the image costs ≈ 10 points on HallusionBench.
  - Seeing and Solving: two-pass extract→solve beats direct answering on charts, but a one-call "state first" prompt does not (contrast with C5).
  - **Our framing:** the sign depends on where the reader stands relative to what the question-blind caption holds.
- Minimal pairs as a design: TempCompass's conflicting videos, MotionBlind, TimeBlind. We apply the pairs to representations, not to models.
- Motion and reversal from pixels: Which Way Did It Move?, MotionBlind, Seeing the Arrow of Time, and the reversal-drop paper. Video models often fail direction, speed and reversal from frames too, which fits P1's null.
- Cost: no paper solves for break-even with a reader downgrade (Q6 open). Related: SAFE-Cascade (per-question routing, −9.3% cost), Sprout (lazy memory), MIT³ (parametric memory), Industrial ReMEmbR (caches vision tokens instead of captioning at write time: the opposite design).
- Visual token budgets: LoHi, VEI.
- Caption utility judged by QA: CaptionQA, CapQuiz, ViSIL.
- Temporal benchmarks: TempCompass, plus T3 (text-to-video temporal transfer; related at most, checked 2026-10-19). Reversed-video shortcuts and the forward prior (find the citation).
- **Still to do:** repeat Q7 in the week before submission; Google Scholar and OpenReview were not searched.

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7. Limitations (bullets)

- One benchmark; short clips; multiple choice only.
- Closed readers run through a CLI: its system prompt is included, and there is no temperature control. Mitigations: pinned IDs, isolation checked on every call, released outputs, and an open-weights reader (C12, Qwen2.5-VL-7B, greedy) that replicates claims 1 and 2.
- Claude reading Claude (C3). Mitigation: the Gemini chain.
- Development → confirmatory drop of Haiku on frames (0.70 → 0.61); about half of it is the mix of video types.
- Questions share filmed clips. Mitigation: clustered CIs.
- Thinking is decided per call by the model (effort medium), so it is observed, not controlled.
- **Question-blind captions drop what the question needs and the captioner did not think to say** (stacked comparisons, reversals). This is a limit of question-blind indexes, not only of our captioners.
- Claims 2a, 2b, 3a, 4, 5, 7 and 8 are exploratory, even though each was predicted in writing before its run (2a was not predicted).

- The audit's judge (Sonnet) also wrote one of the caption sets it judges. Mitigation: the 40 hand labels and κ.
- **54% of the confirmatory videos are manipulated** (reversed, stacked, two-clip), so overall accuracies mix natural video with stress tests. Report plain videos separately (§8.1).

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8. Discussion: why a caption index is hard to beat and hard to evaluate (skeleton)

Core idea. A caption is a question-blind summary shaped by the captioner's prior. Its value lies in where the video departs from that prior, and those departures are:

- what captioners write least;
- what similarity metrics detect least;
- the only places where baselines fail.

8.1 What the benchmark tests when the captioner is also misled

- **The concern:** TempCompass's manipulated videos mislead models. The captioner is a model, so a low score may only mean that the captioner was fooled.
- **Answer:** the system under test is the whole index: a model writes once without the question, and the reader answers later. A captioner that can be misled is part of that system, not a confound. The arms separate the two causes, because the same model also reads the same pixels with the question in view.
- **Table 5** (D3, descriptive; Sonnet as captioner and reader; split of headroom by type; 95% CIs over clip clusters):
  - perception shortfall = 1 − frames: the model fails even when it sees the question, so frames lose this too;
  - index loss = frames − own captions: it saw the answer when asked but did not write it down blind;
  - retention = (captions − question only) / (frames − question only).

Type (n)	Question only	Frames	Own captions	Perception shortfall	Index loss	Retention
plain (137)	0.61	0.89	0.88	0.11 [0.06, 0.16]	+0.01 [−0.03, +0.06]	0.95 [0.79, 1.12]
two-clip (45)	0.29	0.91	0.91	0.09 [0.02, 0.17]	+0.00 [−0.07, +0.06]	1.00 [0.91, 1.12]
reversed (74)	0.18	0.64	0.46	0.36 [0.25, 0.48]	**+0.18** [+0.05, +0.30]	0.62 [0.35, 0.88]
stacked (41)	0.32	0.66	0.39	0.34 [0.17, 0.51]	**+0.27** [+0.09, +0.47]	0.21 [−0.27, 0.67]
all (297)	0.41	0.80	0.71	0.20 [0.15, 0.26]	+0.09 [+0.04, +0.14]	0.77 [0.64, 0.90]

- **Reading:**
  - On natural video (plain), the index keeps almost everything (0.95). The index loss is ≈ 0 on plain and two-clip videos, and its CIs exclude 0 only on reversed and stacked.

    - On reversed videos, the model is misled even when it looks with the question in view (0.36 shortfall; wrong answers equal the prior answer 0.59 of the time, C8). Writing blind loses another 0.18.
    - On stacked videos, the model can see the comparison when asked (0.66), but blind captions keep little of it (0.21; the CI is wide). This is the price of being question-blind (Prop. 4, C5b +0.22).
- **Haiku version** (Haiku reads Sonnet's captions; appendix table): the index loss is negative overall, −0.09 [−0.15, −0.04], so the captions beat Haiku's own frames, mostly on two-clip videos (−0.33 [−0.48, −0.20]); plain −0.05 [−0.11, +0.00], reversed −0.07 and stacked −0.02 (CIs include 0). Haiku's perception shortfall is 0.39 overall and ≈ 0.6 on reversed and stacked. Its retention by type is unstable where its frames are barely above its question-only score (stacked: 0.41 vs. 0.34), so it is not reported by type.
- **Role of the hand labels:** they test whether the judge correctly sorts caption failures into three kinds:
    - misstatement: the captioner was misled and wrote it;
    - omission: the captioner did not commit;
    - reader error.

Table 5 cannot show this split. The hand labels do not split perception from blindness; the arms do that.

- **The captioner prompt is not changed.**
    - The confirmatory prompt is generic by design.
    - A prompt that names the benchmark's tricks (reversal, stacking) tells the index the test, and any gain from it would not generalize.
    - **C11, the middle point:** a whole-video caption prompt with one bullet that names the kinds of question (actions, speed, direction, order, attribute changes) but not the benchmark's tricks.
    - Sonnet on its own captions: 0.721 → 0.761 (C11b +0.040 [+0.003, +0.081]). Haiku: +0.027 (CI includes 0).
    - Sonnet's index loss against its frames halves: −0.077 → −0.037.
    - **Stacked is not fixed.** Against the per-window captions, stacked stays at 0.39, and Sonnet on frames is +0.27 ahead. The apparent stacked gain over the whole-video captions only recovers C10's whole-video loss there.
    - Reversal mentions do not rise (0.05 vs. 0.08 of reversed videos). The captions are 15% longer at the same limit, which may explain part of the gain.
    - **Reading:** knowing the task buys a little; the comparison in stacked videos needs the question itself (C5b +0.22). Question-blind → task-aware → question-aware is a gradient, and stacked sits at its far end.
- The hand-label notes are in §8.3, and κ is in §4.4.

8.2 Why the baselines are hard to beat

- **The prior is strong:** the question alone scores 0.41 for both readers (C8a); 0.61 on plain videos.
- **Errors fall back to the prior:** on reversed videos, Sonnet's wrong answers equal its question-only answer 0.59–0.74 of the time, across frames and caption arms (uniform 0.43; C8). The caption helps only where it contradicts the prior, and blind captioners rarely write that.
- **Omission dominates:** 0.61–0.73 of wrong answers on captions come from captions that are silent on the question (C7). Misstatement is 0.15–0.28 and reader error 0.05–0.17.

- **More pixels do not help the Claude captioner:** C9a +0.003 at 3.6× the visual tokens. The loss is in what gets written, not in what is seen.
- **The value depends on the reader** (Fig. 3): the same index helps Haiku (Sonnet captions: retention 1.47 [1.18, 1.95]) and hurts Sonnet (0.77 [0.64, 0.90]). Reading frames gains 0.19 from Haiku to Sonnet; reading Sonnet's captions gains 0.00. Whether an index "beats the baseline" depends on which reader the baseline uses (claim 2a, the cap).
- **Reconstruction phase (`docs/paper/captions_vs_frames.md`, §4):** continuity baselines were as strong as the models.
  - An 8B LLM in-filling masked captions lost to copying the nearest caption.
  - On forced-choice recognition (160 items), Haiku from captions was level with caption copy (51.2% vs. 45.6%), and frame copy (87.5%) was level with Opus (85.0%).
  - Sonnet's generated captions only drew level with caption copy (text rank 22.3 vs. 23.6).

8.3 Why caption utility is hard to evaluate

- **No question-free reference:** a caption is right or wrong only relative to a question. The same caption is complete for "what happens" and silent for "which copy is faster".
- **Similarity metrics track style:** in the reconstruction phase, cos_sim correlated with caption repetitiveness (APCS_T, Spearman ρ = +0.61; `docs/experiments/evaluation_and_metrics_decision_dossier.md`). Per-second and order-free scoring gave opposite headlines:
  - per-second rank had no headroom (the true caption was level with caption copy, −0.2 ranks);
  - order-free rank showed Claude's fills clearly beating caption copy (`captions_vs_frames.md` §4).
- **"Silent" and "wrong" overlap:**
  - The judge found forward narration in only 0.28–0.45 of reversed failures, against a prediction of ≥ 0.50. A caption that never states a direction is consistent with both readings.
  - The hand labels show the same overlap: all 5 judge/hand disagreements are captions the user left "silent" and the judge decided. The hard items in the user's notes fall into four kinds:
  - a static description that fits a wrong option ("nothing visibly moves" read as "unchanged"; "the upper one holds at 100%");
  - a near-synonym of an option ("places it onto" vs. "dropped");
  - a 1 s last window that contradicts the long first one ("the clip shifts from … to …" in a split-screen video);
  - an answer that needs knowledge of the task: debris "tumbling upward" means "rebuilding" only to a reader who knows clips may be reversed.
- **One caption is one sample:** re-captioning with identical input never gave the same caption (0 of 48), and answers changed on 10–17% of those questions (C10). Only averages over many questions are stable.
- **The judge shares a family with the captioner:** Sonnet judges Sonnet's captions. κ against the hand labels is the check: 0.81.

8.4 What made it measurable

- **Utility through a fixed reader on multiple-choice questions**, not similarity to a reference caption.
- **Controls:**

    - the self-caption control;
    - the reader's own view of the frames;
    - question only.

Without them, a "caption effect" mixes captioner, modality and prior.

- **Minimal pairs:** chance pair accuracy is 0.093, and a reader that falls back to the prior scores 0 on a pair (C4).
- **Reporting by video type:** the overall number hides that the loss is concentrated in two types (Table 5).
- **Many questions per comparison**, because of resampling noise (§4.7).

Appendix (list)

- A. Prompts: caption, reader, describe-then-answer, audit.
- B. The development phase (K0–K2) and how it set the hypotheses; predictions against results (C1–C7).
- C. Amendments, deviations, and the term mapping (D2). Includes the hand-label caveat (C7): the labels were blind to the judge, but made in a session where Claude (Opus) pointed out caption details on 6 items; κ is 0.81 with the first-instinct labels too.
- D. Tables by aspect and by video type; clustered CIs.
- E. The minimal-pair construction (C4), with pair results by group.
- F. Release: code, captions, raw outputs; TempCompass licence (CC BY-NC).

Gaps found while outlining (to decide)

Gap	Why it matters	Fix	Cost
~~40 hand labels~~	Done (2026-10-22): agreement 0.88, κ = 0.81 ≥ 0.6, so Table 4 keeps the judge's labels	—	—
~~Sonnet question-only arm~~	Done (C8): 0.414; claim 5 holds for Sonnet	—	7-day quota 12% → 16% for C8 + C9
~~Claim 3 and the captioner's visual budget~~	Done (C9): 448 px captions do not close the 9 points (C9a +0.003)	—	(above)
~~Window length on short videos~~	Done (C10): whole-video captions are no better (+0.01), and cost half as much per video	—	7-day quota 17% → 19%
~~Discussion §8 numbers~~	Done (D3): Table 5 with clustered CIs, the Haiku version, and the reconstruction-phase numbers cited	—	—
~~Haiku retention CIs~~	Done (D3, §3): CIs exclude 1 for every index but self-captions	—	—
~~C11: captioner prompt that knows the kinds of question~~	Done (2026-10-22): +0.040 (Sonnet), +0.027 (Haiku); halves Sonnet's index loss; stacked not fixed (claim 4a, §8.1)	—	7-day quota 22% → 25%
~~Figure for the reader-strength tension (claim 2a)~~	Done: Fig. 3, a slope chart by reader	—	—
~~Colour consistency~~	Done: Fig. 2 uses palette slots 3–7, which Fig. 1's reader colours don't use; Fig. 3 reuses Fig. 2's caption hues	—	—
Literature: done (N2, no scoop). Repeat Q7 before submission	New work weekly	rerun the §10.3 queries	this session's quota
Public pre-registration archive	Reviewers can't verify local commits	an anonymous repository or OSF, before submission	user
Open-weights reader: **done** (C12, 2026-10-27)	"All readers are closed"	Claims 1 and 2 replicate; 2a at the boundary	Kaggle